From an Uninformative Spectral Bound to Useful Stopping Guarantees for QC-GHP Codes

October 24, 2026

Expanded follow-up to L-F16-stopping-bound; presentation revised

Abstract

Abiad and Castriota ask for QC-GHP parameters making their spectral stopping-set bound informative, and propose stronger spectral bounds. Our earlier universal obstruction answers the original useful-parameter question negatively in the proposition's stated scalar-factor model. Here we show that useful replacements exist: a degree-corrected fourth-moment criterion gives stopping distance at least $\gamma + 1$, and explicit sparse QC-GHP families meet its hypotheses. A fixed column/row-weight 4/8 family has guarantee five; an independently certified 54-variable instance improves on the compared Zhu–Chugg formulas wherever those formulas apply. A two-block incidence refinement and a separate family with exact stopping and logical distance six demonstrate further structural gains. Finally, Gram-equal CSS constructions prove an unbounded limitation for uniformly useful bounds based only on the specified Gram data. The spectral criterion combines classical moment and overlap ingredients; all new proofs, constructions and source-to-matrix bridges are checked in Lean.

1 The source question, its obstruction, and the replacement problem

Aida Abiad and Nichola Castriota pose the motivating question in Section 6 of *Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes*, July 15, 2026, arXiv:2607.13666v1 [1]. Can QC-GHP parameters make Proposition 5.1 give an informative stopping-set bound? Remark 5.2 and Section 6 also motivate tighter spectral bounds.

Throughout, a stopping support of a binary check matrix is a nonempty set of columns for which no row meets exactly one selected column. Its minimum size is the stopping distance $s(H)$. This is a property of the specified check matrix. Logical distance instead minimizes over nontrivial logical words; a lower bound for logical distance alone need not exclude all small stopping supports.

1.1 Why the original parameters cannot provide the desired improvement

The source specializes to a scalar second factor in Section 3.2 and retains that model for Proposition 5.1. Write the expanded real zero–one matrix as $H_X = [A \mid I_r \otimes B^{\mathsf{T}}]$, with $m = r\ell$ rows, positive column weight γ, and constant row weight k. Assume positive first-factor column width, $r, \ell > 0$, and $m \geq 2$ so that the second singular value is defined. Put $\eta = \sigma_2(H_X)^2$, the second largest eigenvalue of $H_X H_X^{\mathsf{T}}$, counting multiplicity. The source requires the strict condition

$$\eta < \gamma/2, \qquad a_{\pm} = \frac{m}{\gamma}\left(1 \pm \sqrt{1 - 2\eta/\gamma}\right).$$

Its large-root conclusion additionally requires excluding the small-size branch: for a stopping support S, one must establish $a_- < |S|$. The spectral inequality alone is therefore not a useful stopping-distance guarantee.

Earlier theorem (original parameter question). *For every binary biregular scalar-factor matrix with these positive dimensions and degrees, it is impossible to have both $\eta < \gamma/2$ and $a_+ > 2$. If the strict spectral condition holds, then $r = 1$, $1 < a_+ \leq 2$, and every stopping support satisfies $|S| \geq 2$ and $a_- < |S|$.*

This is the already checked universal result of [7], not a new theorem of this revision. Its proof separates the two unavoidable regimes. For $r \geq 2$, the repeated scalar block forces $\eta \geq \gamma^2$, contradicting the strict condition. For $r = 1$, binary regularity, edge counting and the Gram trace force the dense regime in which $a_+ \leq 2$ and $a_- < 2$. Positive columns already exclude singleton stopping supports, so the separate small-branch requirement is met but yields no improvement. The all-ones 2×4 example has $\eta = 0$ and stopping distance two: raw spectral-gap examples do exist. The complete proof is `SourceQuestionAnswer.lean` and its frozen `Spectral.lean` dependency; Appendix B gives full current and immutable URLs, including the earlier proof closure.

1.2 What this paper answers next

The obstruction makes the replacement problem substantive: can different spectral information give a stopping bound greater than two on an explicit sparse QC-GHP family? Can incidence information do so beyond the earlier B1 result? We answer both existence questions positively. These are explicit follow-up formulations of the source direction, not a claim to an optimal universal bound or a new numbered conjecture in [1].

Table 1: Source questions, follow-up formulations and exact answers.

Question / source	**Answer and scope**
Abiad–Castriota, 15 July 2026, v1, Prop. 5.1, Remark 5.2, ğ6: useful parameters for the stated bound	Negative, by the earlier universal theorem (Section 1.1).
Same source: tighter spectral stopping bounds, including multi-eigenvalue refinements	YES for useful-replacement existence: a fourth-moment criterion proves $\gamma+1$ on explicit sparse joint Sidon families. Theorem 1 and Section 3 prove existence; Theorem 5 limits uniform improvement on other families.
This requested follow-up: incidence-conditioned replacement exceeding two on a sparse family beyond B1	Positive: Theorems 2–3 give the criterion and families; Theorem 4 gives exact distance six.

The central argument is as follows. Section 2 replaces the failed second-eigenvalue regime by an exact fourth-moment criterion. Section 3 realizes it on actual sparse CSS matrices and compares the resulting guarantees with Zhu–Chugg. Section 4 retains block incidence to sharpen classical overlap counting; Section 5 shows how additional finite incidence information yields an exact distance. Section 6 then proves why the specified Gram data cannot uniformly recover those incidence gains.

1.3 Reference chronology and calendar-time comparison

Table 2 makes the central source dates visible before the proofs. Every elapsed time is measured to this paper's dated revision, October 24, 2026 (Pacific), rather than to a moving reading date. These are source ages, not claims about how long a problem was unresolved. Appendix C records all used sources and held leads, with version distinctions and precision.

Table 2: Central sources and elapsed calendar days to October 24, 2026.

Source / role	Relevant preprint or inspected milestone	Elapsed days
Abiad-Castriota	first and inspected v1: 2026-07-15	101
Kashyap–Vardy (overlap bound)	reported conference presentation: 2003	8333–8697
	reported original composition: 2003-09	8425–8454
Xia-Fu	earliest preprint: 2006-06-12	7439
	inspected arXivv2 stamp: 2006-11-11	7287
Zhu–Chugg (spectral comparison)	conference predecessor metadata: 2005-09-28	7696
	inspected journal version: 2007-12	6872–6902
Schwartz-Vardy	first preprint metadata: 2005-03-23	7885
	inspected journal issue: 2006-03	7512–7542
Wang-Pryadko	first and inspected v1 submission: 2022-03-31	1668
	PDF internal date: 2022-04-01	1667
Kuo-Ouyang	first preprint metadata: 2024-11-20	703
	inspected article-in-press: 2026-03-13	225
Dehghan-Banihashemi	first and inspectedv1: 2019-05-31	2703
	journal volume metadata: 2019	2489–2853
Donovan-Rao-Üsküplü-Yazıcı	first and inspectedv1: 2022-05-01	1637
	journal publication year metadata: 2023	1028–1392
Chee-Ling-Yin	first and inspectedv1: 2010-08-10	5919
	journal issue metadata: 2010-11	5807–5836
Gholami-Samadieh	first and inspectedv1: 2014-05-15	4545
	internal PDF title-page date: 2022-03-28	1671
Amirzade-Sadeghi	first and inspectedv1: 2017-09-04	3337
	internal PDF running draft date: 2018-11-18	2897

For Kashyap–Vardy, the author manuscript is undated; the reported 2003 presentation/composition is distinguished from our inspection. Month- and year-only milestones give ranges rather than invented publication days.

2 Replacing the small spectral gap by a fourth-moment criterion

The failed regime uses only the second singular value. A different spectral quantity can certify a concrete incidence property: distinct columns share at most one check. This property makes a classical stopping-set count useful.

Let H be a finite binary incidence matrix, interpreted over the reals when forming its Gram matrix. Write d_v for its column degrees, r_c for its row degrees, and $\eta_1, \dots, \eta_m$ for all eigenvalues of HH^{T}, counting multiplicity. Thus $M_4 = \sum_i \eta_i^2$ is its fourth singular-value moment. All these quantities refer to the actual expanded matrix.

Theorem 1 (Zero-excess spectral stopping bound)**.** *Suppose every column has degree at least γ and*

$$M_4 + \sum_v d_v = \sum_v d_v^2 + \sum_c r_c^2. \tag{1}$$

Every nonempty stopping support has size at least $\gamma+1$. For a (γ, k)-regular matrix with N columns and m rows, condition (1) *is equivalently*

$$M_4 + N\gamma = N\gamma^2 + mk^2, \qquad \text{or} \qquad M_4 = N\gamma(\gamma + k - 1),$$

using the edge count $mk = N\gamma$. There is no second-eigenvalue gap assumption or separate exclusion of small stopping supports.

Proof. Let q_{uv} be the number of checks common to columns u, v. The real spectral identity gives

$$M_4 = \operatorname{tr}((HH^{\mathsf{T}})^2) = \sum_{u,v} q_{uv}^2, \qquad q_{vv} = d_v, \qquad \sum_{u,v} q_{uv} = \sum_c r_c^2.$$

Consequently the excess in (1) is

$$\Delta := M_4 + \sum_v d_v - \sum_v d_v^2 - \sum_c r_c^2 = \sum_{u \neq v} q_{uv}(q_{uv} - 1). \tag{2}$$

Each term is nonnegative because q_{uv} is a natural number. If $\Delta = 0$, then $q_{uv} \leq 1$ for distinct columns. Choose a column v in a stopping support S. Every one of its at least γ checks must meet another selected column. Each other selected column can account for at most one such check. Hence $\gamma \leq |S| - 1$. □

This criterion combines known ingredients. Abiad–Castriota [1], Lemma 4.8, gives the moment/cycle identity; Kashyap–Vardy [2], Theorem 1, gives the uniform-overlap count, and Xia–Fu [8], Theorem 3, supplies a related minimum-column-weight formulation. The contribution here is the checked spectral corollary, its explicit QC-GHP realizations and the complete source-to-matrix proof chain. `SpectralMoment.lean` and `MomentBound.lean` check the identities and the stopping implication.

A bound valid outside the informative regime. For positive column weight γ, define L from (m, N, γ, k, M_4) to be $\gamma + 1$ when the regular moment equality holds and two otherwise. `MomentReplacement.lean` proves $2 \leq L \leq |S|$ for every nonempty stopping support of every such regular binary matrix. The fallback follows from positive columns excluding singletons. `GlobalMomentFamily.lean` checks that $L = \gamma + 1 > 2$ on the families below for $\gamma \geq 2$. Thus the construction realizes an actual globally valid bound function.

Small positive excess. The same calculation has a quantitative extension. For a natural number μ, if $\Delta < 2\mu(\mu + 1)$, every distinct-column overlap is at most μ: an overlap at least $\mu + 1$ would contribute at least $2\mu(\mu + 1)$ through the two ordered pairs in (2). Covering the neighbors of a selected column then gives

$$\gamma + \mu \leq \mu|S|.$$

`MomentExcess.lean` checks this implication without regularity or an assumed overlap bound. It is a conditional criterion; we do not claim a new infinite sparse QC family realizing its useful positive-excess regime.

3 Explicit sparse families and the useful guarantee

An abstract criterion alone would leave the existence question unanswered. We now construct actual QC-GHP matrices satisfying it, including CSS commutation, degree counts and nontrivial logical content.

3.1 A uniform construction for every even column weight

Fix an even $\gamma \geq 2$ and an odd $n \geq 3^{2\gamma}$. In $\mathbb{Z}/n\mathbb{Z}$ set

$$S = \{3^i : 0 \leq i < \gamma\}, \qquad T = \{3^{\gamma+i} : 0 \leq i < \gamma\}.$$

Let A, B be their binary circulants, with $A_{c,j} = 1$ precisely when $c - j \in S$, and similarly for B, T. Use the scalar-factor convention

$$H_X = [A \mid B^{\mathsf{T}}], \qquad H_Z = [B \mid A^{\mathsf{T}}]. \tag{3}$$

The combined support is Sidon, including repeated summands; all relevant pair sums are less than n. Its two supports are disjoint and distinct columns of H_X meet in at most one check. Circulants commute, so over $\mathbb{F}_2$, $H_X H_Z^{\mathsf{T}} = AB^{\mathsf{T}} + B^{\mathsf{T}}A = 0$. There are n checks and $2n$ variables, column weight γ and row weight 2γ. The actual moment and resulting bound are

$$M_4 = 2n\gamma(3\gamma - 1), \qquad s(H_X) \geq \gamma + 1.$$

Theorem 1 therefore gives an informative stopping guarantee. Both sectors also have nonzero kernel weight at least $\gamma + 1$.

These matrices have nontrivial logical content. The word equal to one on the left block and zero on the right has zero syndrome since γ is even. Its left parity is odd since n is odd, while every companion row, and every linear combination of companion rows, has even left parity. Thus it is outside the stabilizer span. For fixed γ, choosing $n = 3^{2\gamma}(2t + 1)$ gives arbitrarily long matrices of fixed degrees. In particular $\gamma = 2, n = 81$ gives 162 variables and stopping guarantee three. No encoded-dimension formula or positive-rate claim is made. `JointSidon`, `JointMatrices`, `JointPowers`, `JointMomentFamily` and `JointKernel` check these claims.

3.2 Short supports: guarantee five at fixed weights 4/8

The exponential sufficient length is unnecessary for a useful weight-four example. For every odd $n \geq 37$ use

$$S = \{0, 1, 3, 11\}, \qquad T = \{0, 5, 12, 18\} \tag{4}$$

in (3). Then

$$N = 2n, \quad \gamma = 4, \quad k = 8, \quad M_4 = 88n, \quad L = 5.$$

Every nonempty X-stopping support has size at least five; every nonzero kernel word in either sector has weight at least five. The all-ones-left class is nontrivial. Thus a fixed-degree infinite sparse family starts at 74 variables.

To verify the incidence hypotheses, it suffices that each support is separately Sidon and that the mixed sums $s + t$ are unique with their ordered support roles. Disjoint supports and a Sidon union are unnecessary. All relevant natural pair sums are at most 36, so reduction modulo $n \geq 37$ preserves these properties. `SeparateSidon.lean`, `ShortFamilySupports.lean` and `ShortFamily.lean` check the finite identities, no-wrap bridge, actual matrices, moment, both-sector bounds and logical witness. This realizes the girth-at-least-six difference criterion of [1],

Corollary 4.11; the classical criterion receives credit rather than being presented as newly discovered.

The same supports also work modulo 27, by a separate finite certificate rather than the no-wrap argument. `Practical27Core.lean` checks every distinct-column overlap of the actual 54-variable matrix in Lean's kernel. The remaining `Practical27` modules prove $M_4 = 2376$, stopping bound five, both-sector kernel bounds and nontrivial logical content. A search suggested the supports; no external search outcome is a proof premise. Neither this instance nor the general short family is claimed to have exact stopping distance five.

3.3 Comparison with Zhu–Chugg on the certified matrices

Zhu–Chugg [3], Corollary 6, equations 41–42, gives the two expressions

$$\frac{N(2\gamma - \eta)}{\gamma k - \eta}, \qquad \frac{2N(2\gamma + k - 2 - \eta)}{k(\gamma k - \eta)}, \tag{5}$$

where η is the second Gram eigenvalue. Their regular connected setting, strict gap $\eta < \gamma k$ and other applicability hypotheses must be respected.

For the 54-variable instance, take the real check vectors

$$u = \delta_0 + \delta_8 - \delta_4 - \delta_{12}, \qquad v = \delta_{20} + \delta_1 - \delta_{24} - \delta_5.$$

They have disjoint supports, squared norms four, Gram energies 34 and cross energy zero. The two-dimensional Rayleigh certificate therefore gives $\eta \geq 17/2$. Under the required gap $\eta < 32$, the first expression in (5) is negative and the second is at most $297/94 < 4$. The new guarantee five is stronger even after integer rounding. `Practical27Spectral.lean` checks the actual matrix evaluations, eigenvalue implication and formula comparison.

For the joint-support example $\gamma = 2, n = 81$, another exact plane certificate has squared norms 40, energies 280 and cross energy 38. It gives $\eta \geq 121/20 > 6$. Under $\eta < 8$ both expressions (5) are negative, whereas the moment criterion gives three. `JointSpectralBase.lean` checks the comparison. These are exact certificates, not floating-point premises; when a required gap fails, the cited formulas are outside their stipulated regime.

4 Retaining block incidence strengthens overlap counting

The moment criterion certifies overlap at most one. A different useful regime allows larger overlaps between column blocks while retaining smaller overlaps inside each block. Keeping this distinction sharpens the classical count.

Theorem 2 (Two-block incidence bound)**.** *Partition the columns of a finite binary incidence relation into two parts. Let S be a nonempty stopping support. Suppose every selected column has at least γ neighbors, distinct selected columns in the same part share at most σ checks, and selected columns in different parts share at most τ checks, where $\sigma \leq \tau$. Then*

$$2(\gamma + \sigma) \leq (\sigma + \tau)|S|.$$

Either part is allowed to be empty within S. For $\sigma + \tau > 0$, the resulting integer lower bound is $\lceil 2(\gamma + \sigma)/(\sigma + \tau) \rceil$.

Proof. For a selected column v, every incident check must meet another selected column, so

$$\gamma \le \sum_{w \in S \setminus \{v\}} |N(v) \cap N(w)|.$$

Write p, q for the selected counts in the two parts. A selected left column gives $\gamma + \sigma \le \sigma p + \tau q$; a selected right column gives $\gamma + \sigma \le \tau p + \sigma q$. When both parts occur, add the inequalities. If only one occurs, its inequality is $\gamma + \sigma \le \sigma |S|$, which is at least as strong because $\tau \ge \sigma$. □

`PartitionBound.lean` checks the cover, finite sums, partition and one-part cases. The uniform version is $\gamma + \mu \le \mu |S|$. Kashyap–Vardy [2], Theorem 1, assumes uniform column size w and maximum intersection μ, giving $1 + w/\mu$. At $\gamma = 4, \sigma = 1, \tau = 2$, its uniform count gives three, while Theorem 2 gives four.

4.1 Sidon supports verify the required intersections

For a finite abelian group G, call T Sidon if equality $a+b = c+d$ among its elements forces equality of the unordered pairs, including repeated elements. Let $A_{c,j} = 1$ precisely when $c - j \in T$, and set

$$H_X = H_Z = [A \mid A^{\mathsf{T}}].$$

The two column neighborhoods are $j + T$ and $j - T$. Distinct columns in the same part share at most one check. Across parts they share at most two: after fixing one common check c_0, the Sidon identity leaves only c_0 and $v + w - c_0$. Every column has degree $|T|$ and every row has degree $2|T|$. Over $\mathbb{F}_2$, commutation gives $H_X H_Z^{\mathsf{T}} = AA^{\mathsf{T}} + A^{\mathsf{T}}A = 0$. These are the scalar QC-GHP/generalized-bicycle matrices of [1, 4], with $r_1 = n_1 = 1$ and $b = a$. Thus

$$3|S| \ge 2(|T| + 1).$$

`SidonStopping.lean` and `SidonMatrices.lean` include the actual matrix, degree and cardinality bridges, with empty-intersection cases.

4.2 Sparse families for any prescribed fixed guarantee

Theorem 3 (Paired powers family)**.** *Fix an even integer $\gamma \ge 2$ and an odd $n \ge 3^\gamma$. In $G = \mathbb{Z}/n\mathbb{Z}$ take*

$$T = \{1, 3, 3^2, \dots, 3^{\gamma-1}\}, \qquad a(x) = \sum_{i=0}^{\gamma-1} x^{3^i}, \qquad H_X = H_Z = [A \mid A^{\mathsf{T}}].$$

These are binary CSS checks of column weight γ and row weight 2γ, with stopping distance at least $\lceil 2(\gamma+1)/3 \rceil$ and a nontrivial logical class. For fixed γ, the lengths $n = 3^\gamma(2t+1)$ are unbounded while the degrees stay fixed.

Proof. Sort exponents in $3^a + 3^b = 3^c + 3^d$ so that $a \le b$ and $c \le d$. If $b < d$, then the left side is at most $2 \cdot 3^b$, while the right is at least $3^d \ge 3 \cdot 3^b$, a contradiction. The opposite inequality is symmetric; cancel equal largest terms to identify the smaller exponents. All relevant pair sums are below n, since $2 \cdot 3^{\gamma-1} < 3^\gamma \le n$. This proves the Sidon property after reduction and distinctness of the γ marks. Theorem 2 applies with $\sigma = 1, \tau = 2$. The all-ones-left word is a kernel word of odd left parity, whereas every companion stabilizer combination has even left parity. It is therefore a nontrivial logical word. □

`PowersThree.lean`, `PowersFamily.lean` and `SidonMatrices.lean` check the support, no-wrap, matrix and span claims.

Column / row weights	Two-block guarantee	Uniform-overlap guarantee
4/8	4	3
6/12	5	4
8/16	6	5

Choose γ first to obtain any desired fixed guarantee, then increase length with degrees fixed. This does not establish growing stopping distance at fixed degree, positive asymptotic rate or practical parameter efficiency. The sufficient length is exponential in γ; a companion-row stopping support gives the upper witness 2γ, checked in `CSSStopping.lean`.

A second choice, $\gamma = 2d$ and $n = 3^{2d}$, gives stopping and nonzero-kernel weight at least $d + 1$. `GrowingFamily.lean` proves $(2d)3^d \leq n$, so incidence density is at most 3^{-d}; `KernelFamily.lean` supplies the kernel bridge. Here the degree varies with length.

5 Additional incidence information gives exact distance six

The reusable criterion need not be sharp. A separate finite certificate and cover argument can close the gap between it and a logical upper witness.

Theorem 4 (Exact-six family)**.** *For every positive integer t, let $n = 39t$, let A have circulant support $\{0, 1, 4, 6\}$, and set $H_X = H_Z = [A \mid A^{\mathsf{T}}]$. There are $78t$ variables and $39t$ checks, column weight four and row weight eight. Both stopping distance and quantum logical distance are exactly six.*

Proof. The two-block criterion gives four. To exclude sizes at most five, simultaneous translation reduces the base $n = 39$ check to one root in each side. A sound degree-one extension procedure rejects both roots with four further additions. Its exact Boolean tests are reduced in Lean's kernel; no external search result is a premise.

Reduction modulo 39 maps lifted column lists to their base lists, lifts each incident base check, and is injective on each check's neighbor set. Thus a lifted stopping support projects to a base stopping support of no greater size. The checked local-cover theorem transfers the lower bound six to every positive multiple of 39.

For the upper bound, select

$$\mathrm{left}(0),\ \mathrm{left}(-5),\ \mathrm{left}(-4),\quad \mathrm{right}(1),\ \mathrm{right}(5),\ \mathrm{right}(6).$$

These six columns are distinct at every stated lift. Every incident check has a selected partner; an independent exact parity calculation shows all selected check degrees are even, so the word is in the binary kernel. Its left weight is three, while each companion row has left weight four. Left parity therefore separates it from the entire stabilizer span. Every nonzero kernel word has stopping support; the stopping lower bound and this nontrivial six-word give logical distance six in both coinciding sectors. □

`BicycleBaseCore`, `BicycleBase`, `BicycleFamilyCore` and `BicycleFamily` check the finite certificate and cover transfer. `BicycleMatrices`, `BicycleDegrees`, `BicycleLogical` and `KernelStopping` check actual matrices, degrees, syndrome, cardinality and nonstabilizer status. Unlike the earlier B1 witnesses, this upper witness is a logical word rather than a companion row.

It is not assumed elementary: the base reconstruction has selected check degrees zero, two and four. No general formula for encoded dimension is claimed.

For comparison with (5), an exact two-dimensional plane certificate gives $\eta \geq 290/13$ on every 39-cover. The base vectors have disjoint supports, squared norms 13, energies 332 and cross energy 42; the minimum plane quotient is $(332 - 42)/13$. The checked fiber argument scales norms and energies exactly through each cover. Thus both Zhu–Chugg expressions are negative whenever their required gap $\eta < 32$ holds. `SpectralPlane`, `FiberSums`, `BicycleCoverEnergy` and `BicycleSpectralFamily` check this comparison without assuming a floating-point spectral envelope.

6 Why the full check Gram data cannot uniformly recover the gain

The positive results suggest a limitation question: could a universally valid function of the full Gram data recover these incidence gains on every family? The following indistinguishable good/bad pairs obstruct that possibility for the specified inputs.

Theorem 5 (Restricted Gram-data obstruction)**.** *For a constituent circulant A from the paired families above, form*

$$\textit{Good: } H_X = H_Z = [A \mid A^{\mathsf{T}}],$$
$$\textit{Bad: } H'_X = [A \mid A], \qquad H'_Z = [A^{\mathsf{T}} \mid A^{\mathsf{T}}].$$

Both are binary CSS pairs with the same check-matrix dimensions and degrees and the same two separate real check Gram matrices. The bad code has stopping distance two and a nontrivial logical word of weight two. Any real-valued stopping lower-bound function depending only on these shared data and valid on both members is at most two on the good member as well.

Proof. Circulants commute with their transposes. Over the reals,

$$H_X H_X^{\mathsf{T}} = AA^{\mathsf{T}} + A^{\mathsf{T}}A = 2AA^{\mathsf{T}} = H'_X (H'_X)^{\mathsf{T}},$$

and the same calculation gives equality for the Z check Grams. Both pairs commute in the binary CSS sense. The bad X-matrix has identical positive columns at each coordinate, giving a stopping pair; positive columns exclude singletons. The corresponding weight-two word is outside the companion stabilizer span, as checked by the binary parity separation in `BadLogical.lean`. The analogous sector statement is also checked. Validity of a bound on the bad pair forces its value to be at most two; the identical inputs force the same output on the good pair. □

For an unbounded separation, choose $\gamma = 4M$ for $M \geq 1$ and $n = 3^{4M}(2t + 1)$ in Theorem 3. Every good stopping support has size at least $2M + 1$, while its bad twin has stopping distance two. Consequently any such bound L satisfies $ML < |S|$ on every good stopping support. `RealGramBound.lean`, `GramGap.lean` and `BothGram.lean` check the exact quantifiers and both Gram identities; the chain includes `GramObstruction`, `BadStopping`, `SidonBad` and `BadDegrees`.

This obstruction includes functions of the complete Gram spectra and of eigenvectors determined solely by those check Gram matrices. It does not assume equality of binary ranks or encoded dimensions, and it does not include cross-sector real incidence or variable-side eigenvectors as inputs. It is compatible with Theorem 1: the zero-excess criterion succeeds on a particular class, whereas the paired-family obstruction exhibits a different class on which shared Gram data cannot distinguish large from small stopping distance.

7 Relation to classical stopping sets and quantum-code results

The classical ingredients explain both the positive answer and its scope. Kashyap–Vardy [2] already gives a uniform-overlap route to a guarantee above two in the weight-four regime. Theorem 2 improves its value by retaining the partition. Xia–Fu [8], Theorem 3, assumes minimum column weight γ, maximum distinct-column intersection λ and integral γ/λ, and gives pseudo-weight at least $\gamma/\lambda + 1$ with an equality characterization. At $\gamma = 4, \lambda = 2$ this is three, compared with four from the partition criterion. Their girth-at-least-six Theorem 1 does not apply to the paired family's cross-block four-cycles.

Schwartz–Vardy [5] emphasizes that stopping distance belongs to a specified parity-check matrix and that codeword supports are stopping sets. Their stopping redundancy optimizes over added checks, unlike the fixed-H_X guarantees here. Wang–Pryadko [4], equation 8, uses $H_X = [A \mid B]$, $H_Z^{\mathsf{T}} = [B; -A]$; our paired case takes $B = A^{\mathsf{T}}$. Their logical-distance bounds and dimension formula concern different invariants. Likewise, Kuo–Ouyang [6] studies decoding exploiting degeneracy; decoder success does not establish an ordinary peeling stopping-distance bound.

Dehghan–Banihashemi [9] gives cospectral bipartite examples with matched degree sequences and different cycle counts. Those cycle statements explain how spectral data can omit incidence information; they do not supply the actual full-Gram-equal CSS pairs and stopping/logical certificates proved here. The existing source screen is reused without counting a new paper cohort.

Donovan–Rao–Üsküplü–Yazıcı [10], Lemma 9, gives stopping distance at least eight when $3 \nmid a$ for its specified DM/DCA matrices and four-row-group hypotheses. No QC-GHP CSS-companion bridge for our displayed matrices is supplied by that statement. Chee–Ling–Yin [11] distinguishes cyclic difference packing from the partition condition for PCDP; our overlapping supports are not claimed to be a PCDP. Gholami–Samadieh [12] studies classical lifted finite set systems and inevitable walks, and Amirzade–Sadeghi [13] gives classical single/multiple-edge difference criteria and lifting bounds. These comparisons do not make classical stopping bounds above two new. Search coverage and access limitations are recorded once in Appendix D and the accompanying source ledgers.

8 Answers, implications, and remaining targets

The original parameter question is answered negatively by the earlier universal theorem: Proposition 5.1's strict spectral regime cannot produce a large-root guarantee above two in its stated model, with the small branch properly handled. The stronger-replacement existence question is answered positively by Theorem 1 and the explicit sparse families. The incidence-conditioned existence question is answered positively by Theorems 2–3; Theorem 4 additionally supplies exact stopping and logical distance six on a separate infinite family. These are the answers, rather than a report that a search failed or that a raw spectral inequality happened to hold.

The implications have distinct scopes. Full-spectrum moment information is sufficient on a concrete class. Retaining block incidence improves uniform overlap counting. Even the full separate check Gram matrices cannot uniformly recover the incidence gains on the class in Theorem 5. Binary ranks, encoded dimensions and cross-sector data remain outside that obstruction. The short 4/8 family has lower bound five, not an exact-five theorem, and no positive asymptotic rate or growing distance at fixed degree has been established.

A bounded next follow-up is to construct or obstruct informative sparse scalar QC families with positive moment excess, or determine sharp stopping distances for the general powers families. These further questions do not prevent the two existence answers from being complete.

Formal evidence. All new proofs and source-to-matrix bridges are checked in Lean. The unchanged build and audit cover 485 named declarations: 416 new and 69 inherited. Appendix A maps the claims to their complete proof chains, Appendix B gives every full current and immutable Lean URL, and Appendix C gives the complete chronology behind Table 2. Investigation timestamps and inventory reconciliation are supporting records in Appendix D.

References

[1] Aida Abiad and Nichola Castriota. Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes. 15 July 2026, v1, Prop. 5.1, Remark 5.2,ğ6. https://arxiv.org/html/2607.13666v1.

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A Proof coverage and checked Lean excerpts

The aggregate build and all 485 named-declaration audits pass in Lean 4.36.0-rc1 with mathlib `5f7b4198d22a9c82e1c698a379334fe128095c82`. Only propext, Classical.choice and Quot.sound occur. The project contains 50 new modules and four frozen imported research modules. Reproduce with lake build StrongerAnswer, followed by lake env lean AuditStronger.lean. The logs, signatures, complete source links, hashes and independent reconstructions accompany the archive.

Proof component	**Checked source**
Ordered eigenvalue fourth moment, zero excess and useful spectral bound	SpectralMoment; MomentBound
Globally valid bound function and quantitative nonzero excess	MomentReplacement; GlobalMomentFamily; MomentExcess
Shorter fixed-degree family and 54-variable matrix, both sectors	SeparateSidon; ShortFamilySupports; ShortFamily; Practical27Core; Practical27; Practical27Design; Practical27Spectral
Actual source matrices, length/density consumers and bad companion degrees	SourceConsumers; BadDegrees
Constructed joint Sidon matrices, spectral family and both binary sectors	JointSidon; JointMatrices; JointPowers; JointMomentFamily; JointKernel
Exact plane certificates and all 39-cover energy scaling	SpectralPlane; FiberSums; BicycleSpectralBase; BicycleCoverEnergy; BicycleSpectralFamily; JointSpectralBase
Real-valued/unbounded Gram gap and nontrivial bad logical class	RealGramBound; GramGap; BadLogical; BothGram
Growing-degree density and kernel guarantees	GrowingFamily; KernelFamily
Generic neighbor cover and two-block/uniform bounds	PartitionBound.lean
Sidon intersection limits and no-wrap powers-of-three support	SidonStopping.lean; PowersThree.lean
Binary matrices, CSS, degree counts and logical separation	SidonMatrices.lean; PowersFamily.lean
Rooted search soundness, two roots and cyclic covers	dependencies/IncidenceCore,List,Cover; BicycleBaseCore,Base,FamilyCore,Family
Concrete actual matrices, degrees, overlap and exact logical witness	BicycleMatrices,Degrees,Overlap,Logical; KernelStopping
Same-Gram identity, bad pair and universal-bound obstruction	GramObstruction; BadStopping; SidonBad
Known nonzero companion-row stopping mechanism	CSSStopping.lean
Consumers and exhaustive named-declaration audit	StrongerAnswer.lean; AuditStronger.lean

`PartitionBound.lean`

```
theorem partition_bound (color : V → Bool) (S : Finset V)
```

```
    (hS : Incidence.Stopping E S) (γ σ τ : ℕ) (hst : σ ≤ τ)
    (hdeg : ∀ v ∈ S, γ ≤ (neighbors E v).card)
    (hov : ∀ v ∈ S, ∀ w ∈ S, v ≠ w →
      overlap E v w ≤ if color v = color w then σ else τ) :
    2 * (γ + σ) ≤ (σ + τ) * S.card := by
  classical
  let P := S.filter (fun w => color w = false)
  let Q := S.filter (fun w => color w = true)
  have hcard : P.card + Q.card = S.card := by
    have h := card_filter_add_card_filter_not (s := S) (fun w => color w = false)
    simpa [P,Q,Bool.not_eq_false] using h
  have hfalse (v : V) (hv : v ∈ S) (hc : color v = false) :
      γ + σ ≤ σ * P.card + τ * Q.card := by
    have h := column_budget E color S hS γ σ τ hdeg hov v hv
    simpa [hc,P,Q,Bool.not_eq_false] using h
  have htrue (v : V) (hv : v ∈ S) (hc : color v = true) :
      γ + σ ≤ τ * P.card + σ * Q.card := by
    have h := column_budget E color S hS γ σ τ hdeg hov v hv
    simpa [hc,P,Q,Bool.not_eq_true,add_comm] using h
  by_cases hp : P.Nonempty
  · obtain ⟨v,hv⟩ := hp
    have hv' := mem_filter.mp hv
    have hb := hfalse v hv'.1 hv'.2
    by_cases hq : Q.Nonempty
    · obtain ⟨w,hw⟩ := hq
      have hw' := mem_filter.mp hw
      have hb' := htrue w hw'.1 hw'.2
      nlinarith
    · have hq0 : Q.card = 0 := card_eq_zero.mpr (not_nonempty_iff_eq_empty.mp hq)
      nlinarith [Nat.mul_le_mul_right P.card hst]
  · have hp0 : P.card = 0 := card_eq_zero.mpr (not_nonempty_iff_eq_empty.mp hp)
    obtain ⟨v,hv⟩ := hS.1
    have hc : color v = true := by
      cases hc : color v
      · have : v ∈ P := mem_filter.mpr ⟨hv,hc⟩
        exact False.elim (hp ⟨v,this⟩)
      · rfl
    have hb := htrue v hv hc
    nlinarith [Nat.mul_le_mul_right Q.card hst]
```

PowersFamily.lean

```
theorem positive_family (γ n : ℕ) [NeZero n]
    (hn : 3^γ ≤ n) (heven : Even γ) (hodd : Odd n) :
    let S := support γ n
    (SidonStopping.X (ZMod 2) S * (SidonStopping.X (ZMod 2) S)ᵀ = 0) ∧
    (∀ v, (Finset.univ.filter (fun c => SidonStopping.X (ZMod 2) S c v = 1)).card = γ) ∧
    (∀ c, (Finset.univ.filter (fun v => SidonStopping.X (ZMod 2) S c v = 1)).card = 2*γ) ∧
    (∀ T, Incidence.Stopping (SidonStopping.E S) T → 2*(γ+1) ≤ 3*T.card) ∧
    (SidonStopping.X (ZMod 2) S).mulVec (SidonStopping.leftOnes (G := ZMod n)) = 0 ∧
    (¬ ∃ a, (SidonStopping.X (ZMod 2) S)ᵀ.mulVec a =
      SidonStopping.leftOnes (G := ZMod n)) ∧
    (SidonStopping.X ℝ S * (SidonStopping.X ℝ S)ᵀ =
      SidonStopping.BadX ℝ S * (SidonStopping.BadX ℝ S)ᵀ) := by
  dsimp only
  have hcard := support_card γ n hn
  have he : Even (support γ n).card := by simpa [hcard] using heven
  refine ⟨SidonStopping.css _,?_,?_,?_,SidonStopping.leftOnes_kernel _ he,?_,SidonStopping.gram_equal _⟩
  · intro v; rw [SidonStopping.column_weight,hcard]
  · intro c; rw [SidonStopping.row_weight,hcard]
  · intro T hT
    simpa only [hcard] using SidonStopping.stopping_bound _ (sidon γ n hn) T hT
  · apply SidonStopping.leftOnes_not_stabilizer _ he
    simpa using hodd
```

KernelStopping.lean

```
theorem quantum_distance_six (h : 39 ∣ n) :
    (HX (n := n) (ZMod 2) * (HZ (n := n) (ZMod 2))ᵀ = 0) ∧
    (∀ x : Var n → ZMod 2, x ≠ 0 → (HX (ZMod 2)).mulVec x = 0 →
      6 ≤ (BinaryStopping.support x).card) ∧
    (HX (ZMod 2)).mulVec (@word n) = 0 ∧
    (¬ ∃ a : Check n → ZMod 2, ((HZ (ZMod 2))ᵀ.mulVec a) = (@word n)) ∧
    (BinaryStopping.support (@word n)).card = 6 :=
  ⟨css,all_nonzero_kernel_weight_ge_six h,word_kernel h,word_not_stabilizer h,word_weight h⟩
```

SidonBad.lean

```
theorem gram_only_bound (S : Finset G)
    (lower : Matrix G G ℝ → ℕ)
    (valid_bad : ∀ T : Finset (Bool × G), Incidence.Stopping (badE S) T →
      lower (BadX ℝ S * (BadX ℝ S)ᵀ) ≤ T.card) :
    lower (X ℝ S * (X ℝ S)ᵀ) ≤ 2 := by
  rw [gram_equal]
  have h := valid_bad twin (twin_stopping S)
  simpa [twin_card] using h
```

GramGap.lean

```
theorem unbounded_gram_gap (M t : ℕ) [NeZero (liftLength (2*M) t)]
    (lower : Matrix (ZMod (liftLength (2*M) t)) (ZMod (liftLength (2*M) t)) ℝ → ℝ)
    (valid_bad : ∀ U : Finset (Bool × ZMod (liftLength (2*M) t)),
      Incidence.Stopping (SidonStopping.badE (support (4*M) (liftLength (2*M) t))) U →
      lower (SidonStopping.BadX ℝ (support (4*M) (liftLength (2*M) t)) *
        (SidonStopping.BadX ℝ (support (4*M) (liftLength (2*M) t)))ᵀ) ≤ (U.card : ℝ))
    (T : Finset (Bool × ZMod (liftLength (2*M) t)))
    (hT : Incidence.Stopping (SidonStopping.E (support (4*M) (liftLength (2*M) t))) T) :
    (M:ℝ)*lower (SidonStopping.X ℝ (support (4*M) (liftLength (2*M) t)) *
      (SidonStopping.X ℝ (support (4*M) (liftLength (2*M) t)))ᵀ) < (T.card : ℝ) := by
  have hbound := SidonStopping.real_gram_only_bound _ lower valid_bad
  have hrewrite : 4*M = 2*(2*M) := by omega
  rw [hrewrite] at hT
  have hg := arbitrary_fixed_guarantee (2*M) t T hT
  have hgr : (2*M+1:ℕ) ≤ T.card := hg
  have hrr : (2*(M:ℝ)+1) ≤ (T.card:ℝ) := by exact_mod_cast hgr
  have hmul := mul_le_mul_of_nonneg_left hbound (Nat.cast_nonneg M : (0:ℝ) ≤ M)
  linarith
```

BicycleSpectralFamily.lean

```
theorem zhu_chugg_negative (h : 39 ∣ n)
    (hgap : eigs (Family.HX (n := n) ℝ) ⟨1,by exact check_count h⟩ < 32) :
    let s := eigs (Family.HX (n := n) ℝ) ⟨1,by exact check_count h⟩
    (2*(n:ℝ))*(8-s)/(32-s) < 0 ∧
    (4*(n:ℝ))*(14-s)/(8*(32-s)) < 0 := by
  dsimp
  have hs := second_lower h
  have hn : (0:ℝ) < n := by exact_mod_cast NeZero.pos n
  constructor
  · apply div_neg_of_neg_of_pos
    · exact mul_neg_of_pos_of_neg (by positivity) (by linarith)
    · linarith
  · apply div_neg_of_neg_of_pos
    · exact mul_neg_of_pos_of_neg (by positivity) (by linarith)
    · positivity
```

MomentBound.lean

```
theorem regular_spectral_bound (E : V → C → Prop) [DecidableRel E] (γ k : ℕ)
    (hcol : ∀ v, (PartitionStopping.neighbors E v).card = γ)
    (hrow : ∀ c, rowWeight E c = k)
    (hmoment : (∑ i,(StoppingSpectral.eigs (H E) i)^2) + (Fintype.card V : ℝ)*γ =
      (Fintype.card V : ℝ)*(γ:ℝ)^2 + (Fintype.card C : ℝ)*(k:ℝ)^2)
    (S : Finset V) (hS : Incidence.Stopping E S) : γ+1 ≤ S.card := by
  apply spectral_stopping_bound E γ (fun v => by rw [hcol]) _ S hS
  simpa only [hcol,hrow,sum_const,card_univ,nsmul_eq_mul] using hmoment
```

JointMomentFamily.lean

```
theorem fourth_moment (hn : 3^(2*γ) ≤ n) :
    (∑ i,(eigs (JointSidon.X ℝ (leftMarks γ n) (rightMarks γ n)) i)^2) =
      2*(n:ℝ)*(γ:ℝ)*(3*(γ:ℝ)-1) := by
  have hm := StoppingMoment.moment_of_overlap_one
    (JointSidon.E (leftMarks γ n) (rightMarks γ n))
    (JointSidon.overlap_one _ _ (joint_sidon γ n hn) (disjoint_marks γ n hn))
  rw [real_matrix] at hm
  simp_rw [column_degree hn,row_degree hn] at hm
  simp only [sum_const,card_univ,Fintype.card_prod,Fintype.card_bool,ZMod.card,
    Nat.cast_mul,Nat.cast_ofNat,nsmul_eq_mul] at hm
  nlinarith
```

ShortFamily.lean

```
theorem source_stopping_five (hn : 37≤n) (U : Finset (Bool × ZMod n))
    (hU : Incidence.Stopping (fun v c => HX (n := n) (ZMod 2) c v=1) U) : 5≤U.card := by
  refine regular_spectral_bound (E (n := n)) 4 8 (degree hn) (row_degree hn) ?_ U ?_
  · rw [real_matrix,fourth_moment hn]
    simp only [Fintype.card_prod,Fintype.card_bool,ZMod.card,Nat.cast_mul,Nat.cast_ofNat]
    ring
  · simpa only [HX,Incidence.Stopping,JointSidon.binary_incidence] using hU

theorem replacement_value (hn : 37≤n) :
    regularReplacement n (2*n) 4 8 (∑ i,(eigs (HX (n := n) ℝ) i)^2)=5 := by
  rw [fourth_moment hn]
  unfold regularReplacement
  split_ifs with he
  · norm_num
  · exfalso
    apply he
    push_cast
    ring
```

Practical27Spectral.lean

```
theorem zhu_chugg_comparison (hgap : eigs HR ⟨1,by decide⟩ < 32) :
    let s := eigs HR ⟨1,by decide⟩
    54*(8-s)/(32-s) < 0 ∧
      108*(14-s)/(8*(32-s)) ≤ (297:ℝ)/94 ∧ (297:ℝ)/94 < 4 := by
  dsimp
  have hs := second_lower
  refine ⟨?_,?_,by norm_num⟩
  · apply div_neg_of_neg_of_pos
    · nlinarith
    · linarith
  · apply (div_le_iff₀ (by positivity : 0 < 8*(32-eigs HR ⟨1,by decide⟩))).mpr
    nlinarith
```

Dependency guide. PartitionBound provides generic counting. SidonStopping and PowersThree prove incidence hypotheses constructively. SidonMatrices and PowersFamily supply actual binary CSS and nontrivial logical content. BicycleBaseCore/List dependencies give search soundness; BicycleBase rejects small roots; FamilyCore/Cover transfer exact6. BicycleMatrices/Degrees/Logical/KernelStopping check the actual code and logical witness. GramObstruction, BadStopping and SidonBad prove indistinguishability and exact bad stopping2. CSSStopping checks the known stabilizer upper mechanism. AuditStronger checks every named declaration, including every frozen dependency. The second checkpoint adds SpectralMoment/MomentBound for an actual ordered-eigenvalue replacement; JointSidon/Matrices/Powers/MomentFamily/Kernel for constructed useful spectral families; JointSpectralBase for an exact cited-formula comparison; real-valued/unbounded Gram-gap and nontrivial bad logical proofs; exact two-vector certificates with cyclic-cover energy transfer; and growing stopping/kernel guarantees. SeparateSidon/ShortFamilySupports/ShortFamily provide shorter fixed-degree parameters; Practical27Core/Practical27/Practical27Design/Practical27Spectral check the 54-variable example and exact cited-formula comparison. MomentReplacement/GlobalMomentFamily and MomentExcess provide the global bound function and quantitative extension. BothGram covers both separate check Gram matrices, excluding cross-sector/binary-rank information.

B Complete current and immutable Lean URLs

Every URL below is printed in full and clickable. The stronger-bound sources retain the checked revision `bed7719cb3cac4118e4f333690db364566ede03e`. The earlier proof closure retains its own immutable revision.

B.1 This paper: every proof, audit and imported research file

`AuditStronger.lean`
Current: `https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/AuditStronger.lean`
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`BadDegrees.lean`
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`BicycleBase.lean`
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`BicycleCoverEnergy.lean`
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BicycleFamily.lean
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BicycleFamilyCore.lean
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BicycleLogical.lean
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BicycleMatrices.lean
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BicycleOverlap.lean
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BicycleSpectralBase.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleSpectralBase.lean
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BicycleSpectralFamily.lean
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BothGram.lean
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CSSStopping.lean
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FiberSums.lean
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GlobalMomentFamily.lean
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GramGap.lean
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Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GramGap.lean

GramObstruction.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/GramObstruction.lean
Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GramObstruction.lean

GrowingFamily.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/GrowingFamily.lean
Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GrowingFamily.lean

JointKernel.lean
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JointMatrices.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointMatrices.lean
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JointMomentFamily.lean
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JointPowers.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointPowers.lean
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JointSidon.lean
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JointSpectralBase.lean
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KernelFamily.lean
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KernelStopping.lean
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MomentBound.lean
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MomentExcess.lean
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MomentReplacement.lean
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PartitionBound.lean
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PowersFamily.lean
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PowersThree.lean
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Practical27.lean
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Practical27Spectral.lean
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RealGramBound.lean
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SeparateSidon.lean
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ShortFamily.lean
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ShortFamilySupports.lean
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SidonBad.lean
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SidonMatrices.lean
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SidonStopping.lean
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SourceConsumers.lean
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SpectralMoment.lean
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SpectralPlane.lean
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StrongerAnswer.lean
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dependencies/IncidenceCore.lean
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dependencies/Spectral.lean
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B.2 Cited earlier proof closure

Audit.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/Audit.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/Audit.lean

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classification/AuditClassification.lean
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classification/dependencies/Incidence.lean
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source-answer/AuditDenseSourceBridge.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean

source-answer/AuditSourceAnswer.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/AuditSourceAnswer.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/AuditSourceAnswer.lean

source-answer/DenseSourceBridge.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/DenseSourceBridge.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/DenseSourceBridge.lean

source-answer/SourceQuestionAnswer.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean

source-answer/dependencies/Spectral.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/dependencies/Spectral.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/dependencies/Spectral.lean

source-answer/source-ledgers/parent-Audit.lean
Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean
Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean

C Complete source chronology and calendar deltas

This manuscript uses October 24, 2026 in America/Los_Angeles as its dated revision. UTC research/build times are recorded separately. Civil calendar days to 2026-10-24; exact dates give exact differences. Month/year precision yields inclusive minimum/maximum day ranges; dates are not invented. Date ranges are clipped to the revision because these records already existed then. Source age is not unresolved-problem duration.

The central comparison is Table 2; this ledger also includes inherited sources, software snapshots and held leads. Inspection-only milestones do not become publication dates.

Source / role	**Milestone, date and elapsed calendar days**	**Access / precision**
F16: Abiad-Castriota: motivating parameter question; Original question; Prop.5.1 / Remark5.2 / ğ6	first and inspected v1: 2026-07-15; 101 days (day)	inspected full text
F68: Tanner: Minimum Distance Bounds by Graph Analysis; Classical spectral ingredient	earliest inspected manuscript: 2000-07-26; 9586 days (day); journal milestone: 2001; 9063–9427 days (year)	inspected full text
F67: Zhu-Chugg: Bounds on the Expansion Properties of Tanner Graphs; Classical stopping bound; exact Cor.6 comparison	conference predecessor metadata: 2005-09-28; 7696 days (day); inspected journal version: 2007-12; 6872–6902 days (month)	inspected full text
F69: Panteleev-Kalachev: Degenerate Quantum LDPC Codes With Good Finite Length Performance; Scalar-factor construction	first v1: 2019-04-04; 2760 days (day); inspected v3: 2021-11-12; 1807 days (day); journal milestone: 2021-11-22; 1797 days (day)	inspected full text
F70: Leverrier-Zemor: Quantum Tanner codes; Constituent Ramanujan comparison	first v1: 2022-02-28; 1699 days (day); inspected v3: 2022-09-16; 1499 days (day); PDF internal date: 2022-09-19; 1496 days (day)	inspected full text
F71: Feng-Li: Spectra of Hypergraphs and Applications; Classical source cited by Remark5.2; not a proof premise	publication metadata: 1996-09; 10981–11010 days (month)	abstract only full text gap
F72: Li-Sole: Spectra of Regular Graphs and Hypergraphs and Orthogonal Polynomials; Classical comparison lead; not a proof premise	publication metadata: 1996; 10889–11254 days (year)	partial web text archival gap
F73: Prabhu et al.: On the Efficacy of the Peeling Decoder for the Quantum Expander Code; Known stopping-set mechanisms and expansion conditions	first v1: 2025-04-30; 542 days (day); inspected v2: 2025-05-29; 513 days (day)	inspected full text

Source / role	Milestone, date and elapsed calendar days	Access / precision
F74: Wang-Kulkarni-Poor: Finding All Small Error-Prone Substructures in LDPC Codes; Classical rooted narrowing and branch-and-bound	related first preprint: 2006-05-19; 7463 days (day); inspected current-version date: 2009-04-22; 6394 days (day); journal issue: 2009-05; 6355–6385 days (month)	inspected full text
F75: Yao-Gokduman-Pfister: Cluster Decomposition for Improved Erasure Decoding of Quantum LDPC Codes; B1 simulations / peeling-cluster comparison	first and inspected v1: 2024-12-11; 682 days (day)	inspected full text
Connolly2024: Connolly-Londe-Leverrier-Delfosse: Fast erasure decoder for hypergraph product codes; Original stabilizer-support mechanism, credited through inspected F73/F75	first v1: 2022-08-01; 1545 days (day); v3 metadata: 2024-08-21; 794 days (day); journal publication: 2024-08-27; 788 days (day)	metadata only full text not consulted
F76: Prabhu-Gayathri-Kumar: On the Analysis of Stopping Sets Associated with the Quantum Hypergraph Product Code; Later comparison lead; no exact theorem claim	conference metadata: 2026; 0–296 days (year)	full text access gap
RosnesYtrehus2007: Rosnes-Ytrehus: An Algorithm to Find All Small-Size Stopping Sets of Low-Density Parity-Check Matrices; Bibliography-only enumeration lead	conference year: 2007; 6872–7236 days (year)	bibliography only not used as theorem evidence
Morris2025: Kirsten Morris: Analysis of Graph-based Decoders for Quantum Low Density Parity Check Codes; Discovery trail through existing dissertation inventory	repository source version: 2025-08; 419–449 days (month)	discovery record not new full text screen
qLDPCOrg/qLDPC: qLDPCOrg/qLDPC pinned snapshot; Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-23; 1 days (day)	pinned source or infrastructure

Source / role	Milestone, date and elapsed calendar days	Access / precision
Lcrypto/Stopping-sets-enumerator: Lcrypto/Stopping-sets-enumerator pinned snapshot; Software/infrastructure snapshot, not original publication age	inspected commit: 2023-10-29; 1091 days (day)	pinned source or infrastructure
leanprover-community/mathlib4: leanprover-community/mathlib4 pinned snapshot; Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-21; 3 days (day)	pinned source or infrastructure
X-source-announcement: mathCObot source-paper announcement; Accessible X comparison; not a mathematical answer	post UTC date: 2026-07-16; 100 days (day); authenticated browser local displayed post date: 2026-07-15; 101 days (day)	authenticated post detail rechecked final checkpoint
X-earlier-result: A_Lectionary earlier absorbing-set announcement; Earlier repository result; different target	post UTC date: 2026-10-24; 0 days (day)	post inspected in parent audit
other-code-snapshots: Other inspected GitHub mirrors/README-only code; Secondary discovery/control sources; no theorem or earliest-publication claim	recorded inspection: 2026-10-24; 0 days (day)	inspection date only earliest dates unassessed
Lean-runtime: Lean 4.36.0-rc1 runtime; Pinned proof checker; release-age claim not made	recorded use: 2026-10-24; 0 days (day)	use date only release date unassessed
F77: Schwartz-Vardy: stopping distance and stopping redundancy; Classical matrix-dependent stopping convention, codeword mechanism and distinct-column criterion	first preprint metadata: 2005-03-23; 7885 days (day); journal revised manuscript: 2005-12-09; 7624 days (day); inspected journal issue: 2006-03; 7512–7542 days (month)	inspected full text
F78: Wang-Pryadko: Distance bounds for generalized bicycle codes; Binary GB/CSS construction convention and logical-distance comparison	first and inspected v1 submission: 2022-03-31; 1668 days (day); PDF internal date: 2022-04-01; 1667 days (day)	inspected full text

Source / role	Milestone, date and elapsed calendar days	Access / precision
F79: Kuo-Ouyang: Degenerate quantum erasure decoding; Quantum-aware decoding comparison; article-in-press main text screened	first preprint metadata: 2024-11-20; 703 days (day); received: 2025-05-31; 511 days (day); accepted: 2026-02-23; 243 days (day); publication / inspected article-in-press: 2026-03-13; 225 days (day); version of record metadata: 2026-04-29; 178 days (day)	inspected full text
F80: Kashyap-Vardy: Stopping Sets in Codes from Designs; Classical uniform-overlap lower bound, Theorem 1	reported conference presentation: 2003; 8333–8697 days (year); reported original composition: 2003-09; 8425–8454 days (month); inspected revised author manuscript: 2026-10-24; 0 days (day)	full web text screened direct pdf download403 transcription archived
F81: Xia-Fu: Minimum Pseudo-Weight and Minimum Pseudo-Codewords of LDPC Codes; Classical overlap/pseudo-weight comparison, Theorems1/3; full17-page primary screen	earliest preprint: 2006-06-12; 7439 days (day); inspected arXivv2 stamp: 2006-11-11; 7287 days (day); PDF internal title-page date: 2019-05-06; 2728 days (day)	inspected full text
F38: Dehghan-Banihashemi: Cospectral Bipartite Graphs with the Same Degree Sequences but with Different Number of Large Cycles; Earlier cospectral-cycle obstruction; existing23-page source screen reused, no added inventory cohort	first and inspectedv1: 2019-05-31; 2703 days (day); journal volume metadata: 2019; 2489–2853 days (year)	inspected full text
gideonlsx/bicycle-race: gideonlsx/bicycle-race pinned public repository; README/software comparison; not theorem evidence	inspected commit: 2026-10-11; 13 days (day)	public metadata or readme inspected
kywukuo/qLDPC_dec: kywukuo/qLDPC_dec pinned public repository; README/software comparison; not theorem evidence	inspected commit: 2026-01-26; 271 days (day)	public metadata or readme inspected
prior-proof-checkpoint: Cited original-question and B1 proof closure; Earlier checked proof sources, not a new completion	immutable proof-source commit: 2026-10-24; 0 days (day)	checked prior lean sources
new-proof-checkpoint: This stronger-bound proof checkpoint; Checked mathematical source, not a priority date	new research and proof-source checkpoint: 2026-10-24; 0 days (day)	checked lean source

Source / role	Milestone, date and elapsed calendar days	Access / precision
held-lead-1: Generalized Bicycle Codes with Low Connectivity: Minimum Distance Bounds and Hook Errors; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-2: Spectral Theory of Semisimple Bivariate Bicycle Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-3: Logical Operator Decomposition for Distance Analysis of Bivariate Bicycle Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-4: Permutation Decoding and the Stopping Redundancy Hierarchy; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-5: Quantum Maxwell Erasure Decoder for qLDPC codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-6: Stopping Sets of Algebraic Geometry Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-7: On the Hardness of Approximating Stopping and Trapping Sets in LDPC Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-8: A class of LDPC codes based on symplectic space over finite fields; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed

Source / role	Milestone, date and elapsed calendar days	Access / precision
F82: Donovan-Rao-Üsküplü-Yazıcı: QC-LDPC Codes from Difference Matrices and Difference Covering Arrays; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2022-05-01; 1637 days (day); journal publication year metadata: 2023; 1028–1392 days (year)	inspected full text
F83: Chee-Ling-Yin: Optimal Partitioned Cyclic Difference Packings; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2010-08-10; 5919 days (day); journal issue metadata: 2010-11; 5807–5836 days (month)	inspected full text
F84: Gholami-Samadieh: QC-LDPC Codes Based on Finite Set Systems; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2014-05-15; 4545 days (day); internal PDF title-page date: 2022-03-28; 1671 days (day)	inspected full text
F85: Amirzade-Sadeghi: Lower bounds on the lifting degree by difference matrices; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2017-09-04; 3337 days (day); internal PDF running draft date: 2018-11-18; 2897 days (day)	inspected full text
postlemma-held-1: Difference-Spectrum Completion of Multi-Weight Optical Orthogonal Codes; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	publisher429 and primary PDF403; fulltext unavailable; metadata only no zero question claim
postlemma-held-2: On the Girth of Tanner QC-LDPC Cycle Codes: An Algebraic Number Theory Approach; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	publisher429 and primary PDF403; fulltext unavailable; metadata only no zero question claim
postlemma-held-3: Cyclic balanced incomplete block designs with block size four and arbitrary index; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	publisher403; abstract metadata only; fulltext unavailable no zero question claim

Source / role	Milestone, date and elapsed calendar days	Access / precision
postlemma-held-4: Additive, Structural, and Multiplicative Transformations for the Construction of Quasi-Cyclic LDPC Matrices; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	Secondary search snippet only; primary full version not located/inspected in bounded search; no theorem premise or zero-question claim
postlemma-held-5: Cyclic Difference Packing and Covering Arrays; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	Secondary meta-data/bibliographic lead only; primary full version not inspected; no theorem premise or zero-question claim
X-final-search: Final authenticated X source/stopping audit; Read-only bounded query with known-positive control; not absence or theorem evidence	final inspection: 2026-10-24; 0 days (day)	authenticated bounded search read no post sent

Inspection-only dates are explicitly distinguished from publication dates. The Kashyap-Vardy author manuscript is undated after its reported 2003 composition; its web transcription hash is not a PDF hash. Supplementary/final-version, citation-index and X access gaps remain named. Date differences are source ages, not durations of unsolved problems.

D Search coverage, inventory and revision record

Public GitHub comparisons include the pinned bicycle-race README, which describes logical-distance records, and the Kuo decoder repository metadata. A final CLI support-fingerprint query and a known-positive mathlib control both returned no items; those negative code-search outputs are discarded. Authenticated Edge access later succeeded: the arXiv-ID query visibly returned the source announcement and an earlier project lift-label post. The source post detail displayed10:23 PM, 15 July 2026; the prior UTC 16 July date is distinguished in the chronology. A focused QC-GHP stopping query displayed no results in that bounded view. None is a proof of global absence or priority. The earlier Mac-locked failure is superseded by this fresh read-only check. No post was sent.

Nine additional accessible main texts were screened, adding 28 historical questions/directions, nine precise enough for the denominator:267→295 entries,135→144 eligible targets,110→119 papers and 97→106 full-text main-paper screens. The separate indexing deltas are20/8/4,3/1/1 and 5/0/4. Completed canonical Lean targets remain five. New harvested questions have unassessed current resolution, Lean discovery and effort. The latest four-paper target screen records five broad directions; FSS and the lifting-degree paper have no additional explicit unresolved target. The PCDP theorem's excluded residue classes are not invented as an open conjecture: explicit 8/16/32 seeds exist in the same paper. Existing F38 screening is reused without duplication. Publisher429/403, supplementary/version, retrieval and X access gaps remain explicit.

Research time versus manuscript revision. The stronger-bound investigation began October 24, 2026 at 22:59:20 UTC. The first checked incidence/exact-six checkpoint was reached at 23:36:01 UTC (36 minutes 41 seconds). The final selected proof checkpoint was October 25 at 01:30:24 UTC (2 hours 31 minutes 4 seconds). These timestamps describe the completed mathematical investigation. The present LaTeX and narrative revision changes presentation and preserves its checked proof sources, inventory totals and immutable proof pins. Its publication timestamp is recorded separately in the accompanying presentation-revision record. The manuscript's civil revision date is October 24, 2026 in America/Los_Angeles; UTC timestamps are distinct.